Taining An

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RESEARCH INTERESTS

Human exercise interventions and physiological adaptations to training; effects of exercise and dietary interventions on physical fitness, body composition, and metabolic health; relationships between changes in adiposity and systemic inflammation.

EDUCATION

Beijing Sport University Beijing, China
M.S. in Sports Training GPA: 90/100 *Sep* 2024 – *Jun* 2026
Thesis: Effects of Time-Restricted Eating Combined with High-Intensity Functional Training and Aerobic Exercise on Systemic Immune-Inflammatory Status and Body Composition in Young Adults with Obesity

Chengdu Sport University Chengdu, China
B.S. in Sports Training GPA: 3.21/4 (Junior–Senior: 3.48/4.0) *Sep* 2019 – *Jun* 2023

RESEARCH EXPERIENCE

Effects of Exercise and Time-Restricted Eating on Body Composition and the Inflammatory Profile of Young Adults with Obesity: A Four-Arm Randomized Controlled Trial
Master's Thesis Research | First-Author Manuscript in Preparation
Beijing Sport University Advisor: Prof. Yanchun Li *Sep* 2025 – Present

- Designed and led the day-to-day management of an 8-week, four-arm randomized controlled trial (N = 122) comparing supervised high-intensity functional training plus aerobic exercise, time-restricted eating with an 8-hour eating window, their combination, and a no-intervention control.
- Oversaw participant screening, recruitment, randomization, supervised exercise delivery, and adherence monitoring, using heart-rate and RPE data to guide training progression and daily dietary logs to verify TRE adherence.
- Conducted DEXA body-composition assessments and collected and processed fasting venous blood samples; systemic inflammation was evaluated using SII, IL-6, and TNF-α, with cytokine concentrations determined by automated chemiluminescent immunoassay.
- In the completed three-arm thesis subset (n = 65), the exercise group showed a significant reduction in lnSII ($P < 0.001$); changes in fat mass and lnSII were not significantly correlated ($r = 0.20$, $P = 0.12$).
- Conducting full-cohort analyses of IL-6 and TNF-α using ANCOVA, hierarchical regression, dose-response modeling, and bootstrap mediation to evaluate whether fat-mass change mediates exercise-related changes in inflammation.

VO_2 max Treadmill Validation (Industry Collaboration with Xiaomi)
Beijing Sport University Advisor: Prof. Yanchun Li
Research Assistant *Mar* 2025 – *May* 2025

- Personally administered incremental-load treadmill VO_2 max tests on 150+ elite runners (cohort total: 230) to validate Xiaomi smartwatch readings against Garmin reference devices.
- Managed the full testing protocol hands-on: fitting gas-analysis masks, controlling treadmill ramp progression, and monitoring real-time gas-exchange data via a Cortex metabolic analyzer.
- Contributed validation data to Xiaomi's device calibration and algorithm refinement.

Biochemical Laboratory Research: PBMC Isolation & DNA Extraction
Beijing Sport University Advisor: Prof. Yanchun Li & Prof. Zihong He
Research Assistant *Sep* 2024 – *Jul* 2026

- Processed 154 venous blood samples for studies on exercise-induced physiological adaptation.
- Performed PBMC isolation by density-gradient centrifugation and DNA extraction using magnetic bead-based purification.

- Assisted with force platform testing (pre- and post-intervention, 260+ participants) and ultrasound imaging data collection for concurrent studies in the laboratory.

PUBLICATIONS

PEER-REVIEWED PUBLICATIONS

Nan, L., Wu, M., **An, T**., Li, Y., & Yang, X. (in press). Effects of time-restricted eating and concurrent training on body composition and dietary intake in young adults with obesity: A four-arm randomized controlled trial. *Journal of Exercise Science & Fitness.*

Yan, H., Zhu, Y., Chen, H., Chen, L., Ren, W., Guo, Q., Guo, Z., Dong, H., Yang, J., Xu, Q., & **An, T**. (2026). Temporal structure of national vs. world-level taekwondo competitions: A time-motion analysis in taekwondo. *Frontiers in Sports and Active Living*, 8, 1897013. https://doi.org/10.3389/fspor.2026.1897013

MANUSCRIPT IN PREPARATION

An, T., et al. Does Change in Fat Mass Mediate Exercise-Induced Changes in the Inflammatory Profile of Young Adults with Obesity? A Mediation Analysis of a Four-Arm Randomized Controlled Trial. Manuscript in preparation.

CONFERENCE PRESENTATIONS

An, T., & Li, Y. (2025). Impact of integrative neuromuscular training on movement accuracy and technical performance in BMX athletes. *Conference presentation at the 2025 National Convention on Sports Science of China.*

EXERCISE INTERVENTION & TEACHING EXPERIENCE

Physical Education Instructor (Intern)
China Agricultural University, Department of Physical Education *Sep* 2025 – *May* 2026

- Designed and delivered, under faculty supervision, an eight-week supervised HIFT and aerobic exercise program for 140 university students across four classes.
- Coordinated six 90-minute sessions per week, maintaining consistent program delivery and monitoring attendance and adherence.
- Individualized and progressed exercise loads using real-time heart-rate and Borg CR10 RPE monitoring, increasing aerobic intensity from 50–60% to 75–85% HRmax; 138 of 140 students completed the program.

SKILLS

- Clinical research: Randomized controlled trials coordination; participant recruitment and randomization; supervised exercise interventions delivery; dietary-log verification; Intervention adherence monitoring
- Physiological and laboratory assessment: DEXA; VO_2 max testing; fasting blood processing; PBMC isolation; DNA extraction
- Statistics analysis: Repeated-measures ANOVA; ANCOVA; hierarchical regression; effect-size estimation; mediation analysis
- Software: R; SPSS; G*Power; PROCESS Macro
- Languages: Chinese (native); English (TOEFL iBT 100)

AWARDS & HONORS

- Graduate Scholarship (First-Class), Beijing Sport University (2025)
- Graduate Scholarship (First-Class), Beijing Sport University (2024)
- Outstanding Graduate, Chengdu Sport University (2023)
- National First-Class Volleyball Player, China
- National First-Class Volleyball Referee, China